

Research paper

Serotonergic multilocus genetic variation moderates the association between interpersonal relationship and adolescent depressive symptoms[☆]

Zihao Zeng^{a,b}, Liyi Peng^a, Shuangjin Liu^a, Qin Yang^a, Hongcai Wang^a, Zhen He^a,
Yiqiu Hu^{a,c,d,e,*}

^a School of Educational Science, Hunan Normal University, Changsha 410081, China

^b Department of Clinical Psychology, Vrije Universiteit Amsterdam, Van der Boerhorststraat 7, 1081 BT Amsterdam, the Netherlands

^c Research Center for Mental Health Education of Hunan Province, Changsha 410100, China

^d Cognition and Human Behavior Key Laboratory of Hunan Province, Changsha 410081, China

^e Center for Mind-Brain Science, Hunan Normal University, Changsha 410081, China

ARTICLE INFO

Keywords:

Depressive symptoms
Interpersonal relationship
Serotonin
Insomnia
Genetic-environmental
THP2 gene

ABSTRACT

Background: Research suggests that genetic variants linked to serotonin functioning moderate the association between environmental stressors and depressive symptoms, but examining gene–environment interactions with single polymorphisms limits power.

Methods: A multilocus genetic profile score (MGPS) approach to measuring serotonergic multilocus genetic variation and examined interactions with interpersonal relationship, insomnia with depressive symptoms as outcomes in an adolescent sample (average age = 14.15 ± 0.63 years since first measurement; range: 13 to 15). **Results:** (1) interpersonal relationship predicted adolescent depressive symptoms; (2) insomnia mediated the effect of interpersonal relationships on adolescent depressive symptoms; (3) the THP2 gene rs4570625 polymorphism G allele was a key risk factor for depressive symptom, and the MGPS moderated the effects of teacher–student relationship and insomnia on adolescent depressive symptom. Specifically, as the MGPS increased, the effects of insomnia on adolescent depressive symptom were enhanced; further, when the MGPS score increased, the effect of teacher–student relationship on depression showed a similar phenomenon with an increased slope and enhanced prediction; and (4) the results of sensitivity analysis showed that multilocus genetic interaction with the environment had a better explanatory power and stability for depression than single polymorphism studies.

Conclusion: MGPS provides substantial power to examine gene–environmental interactions linked to affective outcomes among adolescents.

1. Introduction

Depression, a mental illness that severely limits psychosocial functioning and reduces quality of life, causes the third highest economic burden and medical stress of disease globally and is projected to rank first by 2030 (Malhi and Mann, 2018). Adolescents have a higher risk of depression due to their incomplete psychological development and low mental capacity to tolerate and adjust (McKetta and Keyes, 2019). Adolescent depressive symptoms not only impacts academic performance and social adjustment but also increases the risk of substance use

and suicide (Johnco and Rapee, 2018). Adolescent depressive symptoms also raises the risk of developing related psychiatric disorders in adulthood (Weersing et al., 2017). In addition, social isolation due to the COVID-19 pandemic puts adolescent who rely heavily on interpersonal relationships for emotional support and social development at higher risk for depression and anxiety (Magson et al., 2021).

The interpersonal theory of depression emphasizes the important role of interpersonal relationship in an individual's psychological development. Positive and healthy interpersonal relationship can significantly improve individual's social adjustment and well-being.

[☆] Author Note: The authors have no known conflict of interest to disclose. This article was facilitated by a scholarship of Hunan Education Science Key Funding Project (grant numbers XJK23AJD023) to Dr. Y. Hu and Postgraduate Scientific Research Innovation Project of Hunan Province (CX20210425) to Shuangjin Liu.

* Corresponding author at: School of Educational Science, Hunan Normal University, Changsha 410081, China.

E-mail address: fortune2000947@163.com (Y. Hu).

<https://doi.org/10.1016/j.jad.2023.08.085>

Received 22 March 2023; Received in revised form 7 August 2023; Accepted 14 August 2023

Available online 18 August 2023

0165-0327/© 2023 Published by Elsevier B.V.

Conversely, negative interpersonal relationships, especially dangerous and hostile ones, cause individuals to develop negative cognitive schemas and self-evaluations, thus leading to a higher risk of psychological crisis (Stewart and Harkness, 2017).

In an autobiographical overview of risk factors for depression, Hammen (2018) noted that compared to positive, accepting and warm interpersonal relationship, individuals who experienced more rejection, exclusion and conflict in interpersonal interactions developed negative self-evaluations and cognitive schemata and had a higher risk of depression. Furthermore, Gorostiaga et al. (2019) found that parental warmth was negatively associated with adolescent depressive symptoms. In contrast, parental control was associated with anxiety, depression and suicidal ideation. Moreover, as adolescents grow older, they become more willing to get along with their peers, and peer relationship becomes the primary source of social support (Clayborne et al., 2019). Therefore, previous research found that safe and trustworthy peer relationship improved positive resources for adolescents (Zimmer-Gembeck et al., 2009), while conversely, peer aggression led to depression and anxiety (Stapinski et al., 2015).

The teacher-student relationship is the relationship between students and teachers and is one of the most important interpersonal relationships in school (Longobardi et al., 2021). Harmonious teacher-student relationship increases students' motivation and interest in learning and encourages children to form good multiple emotional connections with family and peers (Dong et al., 2021). Moreover, in a randomized controlled trial exploring the effect of positive intervention on school relationship and depression, the results showed that positive intervention improved teacher-student relationship and reduce the risk of depression by reducing peer alienation (Van Loon et al., 2019). Therefore, the current study proposes to investigate the effect of interpersonal relationship on adolescent depressive symptoms and proposes *hypothesis 1*.

H1. Interpersonal relationship predicts adolescent depressive symptoms

1.1. The mediating role of insomnia

Good sleep is one of the necessary conditions for physiological and psychological functions (Hosker et al., 2019). Compared with adults, adolescents need adequate sleep for healthy physical and intellectual development, while Chinese adolescents commonly suffer from sleep deprivation, nightmares, insomnia and other related problems (Li et al., 2018). Related studies have found that insomnia not only causes short-term consequences such as poor concentration and lower academic performance (Ahrberg et al., 2012) but also may impair brain function in children in the long term, leading to emotional-behavioral problems such as depression and anxiety (Palmer and Alfano, 2017; Tempesta et al., 2018).

In addition, in a review exploring the role of sleep in depression development, Pandi-Perumal et al. (2020) found that insomnia may contribute to depressive symptoms by affecting circadian rhythms, the body's melatonin levels and neuroinflammation. Furthermore, a study showed a strong correlation between interpersonal relationship and sleep quality, and individuals with more interpersonal distress had a higher level of self-awakening and rapid eye movement percentages during sleep (Gunn et al., 2014). Moreover, Gordon et al. (2021) found that the quality of interpersonal relationship could significantly affect the quality of sleep. Specifically, safe relationship improved the quality of sleep, and conversely, negative and hostile relationships led to insomnia. Meanwhile, a study on the relationship between perceived academic pressure and depression among Chinese adolescents showed that perceived academic pressure could affect adolescent depressive symptoms through insomnia (Zhang et al., 2022). In view of this, this study proposes *hypothesis 2*:

H2. Insomnia mediates the effect of interpersonal relationship on adolescent depressive symptoms.

1.2. Effect of serotonergic multilocus on depression

Throughout the past decade of research, the mechanism of interaction between genetic polymorphisms and environmental differences on psychopathology, especially depression, has received extensive attention (Hu et al., 2022; Zeng et al., 2023). Compared to a single polymorphism-environment design with lower explanatory power and higher false positive results (Januar et al., 2015; Starr et al., 2021; Starr and Huang, 2019), studies have found that genetic risk is cumulative, and polygenes have a greater cumulative effect size as well as stronger predictive power (Bulik-Sullivan et al., 2015; Hou et al., 2023; Vrshek-Schallhorn et al., 2019). In a study exploring the effect of five serotonin system polymorphisms and interpersonal stress on depression in 387 youth, Vrshek-Schallhorn et al. (2015) found that the interaction of cumulative polygenic score and interpersonal stress affected depression after controlling for other stressful events. Overview of related literature, dysregulated function of monoamine neurotransmitters such as serotonin and dopamine is the main cause of depression (Andrews et al., 2015; Barton et al., 2008; Gong et al., 2017). Serotonin, also known as 5-hydroxy tryptamine (5-HT), is a monoamine neurotransmitter that regulates a variety of physiological functions and pathological states in the body, such as eating, sleeping, pain and emotions. A recent study found that serotonergic multilocus genetic profile score (MGPS) contributed to sensitivity to social threat, then leading to increased depression (Vrshek-Schallhorn et al., 2022).

Overview of related literature, the diathesis-stress model and the differential susceptibility model are the mainstream theories. The former states that individuals carrying undesirable gene are more likely to have psychological and behavioral problems when they are in stressful or high-pressure situations, while they are no different from normal individuals in positive environments. Different susceptibility model proves that the same genotype can make individuals better or worse in different environments (Zeng et al., 2023; Zhang et al., 2016). This means that individuals with certain genes are at higher risk for psychological problems and disorders in negative environments, but also perform better in positive environment. Therefore, this study proposes to select serotonin system candidate genes to examine the effect on adolescent depressive symptoms.

Among the many candidate genes of the 5-HT system, this study selected tryptophan hydroxylase 2 (TPH2, the rate-limiting enzyme that regulates 5-HT synthesis), SLC6A4 (an important gene encoding the serotonin transporter), the 5-HT 1A/2A receptor (5-HTR1A/2A) and monoamine oxidase A (MAOA which encodes a gene involved in hydrolysis).

1.2.1. Serotonin synthesis: THP2 gene rs4570625 and rs4290270

The interaction between the THP2 gene rs4570625 and environmental stress was found to predict depression in women (Mandelli et al., 2012). Moreover, in a study investigating the correlation between the TPH2 gene and depression in 185 patients with major depressive disorder (MDD) and 64 healthy controls, rs4290270 polymorphic T locus carriers exhibited neural mechanisms of circadian rhythm disturbance, and Tao et al. (2018) suggested that rs4290270 may serve as a reliable biomarker to identify MDD.

1.2.2. Serotonin transport: SCL6A4 gene rs1042173

The SLC6A4 gene rs1042173 is an important candidate for depression. Previous studies have shown that the SLC6A4 gene rs1042173, rs25533 and age can explain 9 % of physical resilience and that the SLC6A4 gene rs1042173 polymorphism is significantly associated with depression (Resnick et al., 2015). Nonen et al. (2016) found that rs1042173 was significantly associated with 5-hydroxytryptamine and serotonin-norepinephrine reuptake inhibitor (SSRI/SNRI) treatment

efficacy at 6 weeks. In addition, in a study exploring the relationship between alcohol craving and SLC6A4, individuals with the rs1042173 polymorphism TT genotype had a more potent need to drink alcohol (Ait-Daoud et al., 2012).

1.2.3. Serotonin receptor: 5-HTR1A gene 1364043 and rs6295

In a study of the association between 5-HT1A gene polymorphisms and schizophrenia in a Chinese Han population, the results showed that the 5-HT1A gene rs10042486 and rs1364043 CT haplotypes showed significant differences between schizophrenia cases and healthy controls (Zhang et al., 2016). Moreover, Wang et al. (2015) showed that carrying the 5-HTR1A gene rs6295 was significantly associated with depression.

1.2.4. Serotonin receptor: 5-HTR2A gene rs7997012 and rs6311

In genetic association analysis studies, the 5-HTR2A genes rs6311, rs7997012, and rs6313 were found to be possibly associated with treatment response to some antidepressants in MDD (Rybakowski, 2012; Zhang and Stackman Jr, 2015). In a review on the genetic mechanisms of 5-HT receptors and depression, Fabbri et al. (2013) found an association between the 5-HT2A gene rs6311 polymorphism A allele and depression.

1.2.5. Serotonin metabolism: MAOA gene rs6323

The MAOA gene rs6323 polymorphism G allele transcribes a higher MAOA enzyme activity, while the T allele transcribes a lower MAOA enzyme activity (Hotamisligil and Breakefield, 1991; Jansson et al., 2005). It is highly likely that the highly active G allele, by encoding a highly active MAOA enzyme, accelerates the metabolism of monoamine neurotransmitters, thereby decreasing the levels of monoamine transmitters in the brain and subsequently, thus increasing the risk of depression (Larson et al., 2010). Therefore, the current study proposes the hypothesis 3:

H3. Serotonergic multilocus genetic profile score moderates the effects of interpersonal relationship and insomnia on adolescent depressive symptoms.

1.3. Current study

In summary, based on the interpersonal theory of depression, the current study uses the polygenic-environmental research paradigm of depression to explore the effects of interpersonal relationships, insomnia, and the polygenic serotonin system on adolescent depression by building a moderating mediation model (Fig. 1).

2. Method

2.1. Participants and procedure

The participants were obtained from junior high school in Hunan,

China. The whole group sampling was taken by class, and informed consent was obtained from the school, students and their parents. The first wave was on Tuesday and Wednesday afternoons of the third week of May 2022, measuring the independent variable (interpersonal relationship) and the baseline dependent variable (adolescent depressive symptoms). A total of 760 data points were collected, including 415 (54.6 %) girls and 345 (45.4 %) boys; 344 (45.2 %) were only children and 416 (54.8 %) were not only children; parents' marital status: 619 (81.5 %) were first married, 38 (5.0 %) were divorced, 45 (6.5 %) were remarried, and 53 (7.0 %) were other types; family economic level: 141 (18.5 %) were very good, 441 (58.0 %) were good, 166 (21.9 %) were average, 8 (1.1 %) were poor, and 4 (0.5 %) were very poor.

The second wave measured the mediating variable (insomnia) on Tuesday, Wednesday and Thursday afternoons of the second week of November 2022. The questionnaire measurement process was the same as in the first wave. As genetic data (oral epithelial cells) were required at this time, the head teacher informed the students before collection that each student should use water to rinse their mouth. A total of 701 gene samples were obtained (as some students refused to participate). Samples were sent back to the biotech company using ice bags immediately after collection. Subsequently, DNA was extracted and purified. All DNA was examined, and the results showed that they were all approximately 10 ng/μl, meeting the requirements of iMLDR multiplex SNP typing technology. The main quality control procedures included a reproducibility check (5 %), negative control check and secondary allele frequency check. The third wave was postponed due to the epidemic until February 2023 (Wednesday and Thursday afternoons of the third week) to measure the dependent variable (adolescent depressive symptoms), and the measurement procedure was the same as before.

A total of 525 participants (average age = 14.15 ± 0.63 years since first measurement; range: 13 to 15) were subsequently matched with two follow-up and genetic data, including 236 (45.0 %) girls and 289 (55.0 %) boys; 250 (47.6 %) were only children and 275 (52.8 %) were not only children; parents' marital status: 436 (83.0 %) were first married, 21 (4.0 %) were divorced, 37 (7.0 %) were remarried, and 31 (5.9 %) were other types; family economic level: 93 (17.7 %) were very good, 316 (60.2 %) were good, 105 (20.2 %) were average, 6 (1.1 %) were poor, and 5 (1.0 %) were very poor. The demographic characteristics of the sample differed significantly only in terms of gender between the first measurement and final sample ($\chi^2_{\text{gender}} = 11.58, p < .001$; $\chi^2_{\text{only-child}} = 0.69, p = .41$; $\chi^2_{\text{parents' marital status}} = 1.92, p = .58$; $\chi^2_{\text{family economic level}} = 1.69, p = .80$).

Starr et al. (2014) stated that the effect size (r^2) of gene–environment interaction generally reached the level of significance at 0.01 to 0.02 ($\alpha = 0.05$). Based on the effect size (0.01, 0.02), the analysis was performed using G*Power 3.1.9.7 software, and to satisfy the large effect volume of Cohen $d > 0.8$, approximately 395–787, would be needed. The sample in this study was 525, which met the basic requirements of the above sample size.

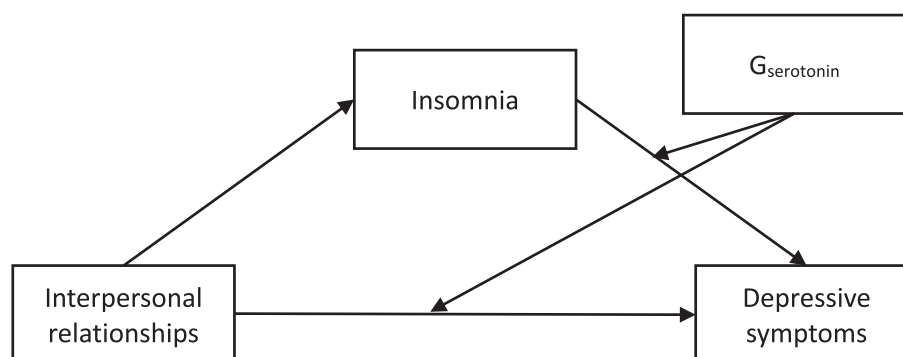


Fig. 1. Theoretical hypothesis model.

2.2. Instruments

2.2.1. Adolescent depressive symptoms

The Short Form of the Center for Epidemiological Studies Depression Scale (SF-CES-D), developed by [Andresen et al. \(1994\)](#), had good reliability and validity in Chinese adolescents ([Yang et al., 2018](#)). The scale consists of 10 items (e.g., *I have no motivation to work or study*) and is rated on a four-point scale (0–3), ranging from 0 for “none or very little” to 3 for “most or all”, with a higher score indicating more severe depressive symptoms. Ten participants were randomly selected to evaluate the questionnaire after investigation, and none of them had any serious discomfort. A study has demonstrated that Guttman’s lambda coefficients have better lower limits of reliability estimation than Cronbach’s α ([Sijtsma, 2009](#)), and the Guttman’s lambda coefficients for adolescent depression in this study were $T1 = 0.76$ and $T3 = 0.85$, with better reliability than 0.7. Fit indices for the confirmatory factor analysis: $T1: \chi^2 = 139.23, df = 35, CFI = 0.90, TLI = 0.88, RMSEA = 0.08, SRMR = 0.05; T2: \chi^2 = 326.30, df = 35, CFI = 0.87, TLI = 0.84, RMSEA = 0.13, SRMR = 0.08$, and the validity of the scale structure was good.

2.2.2. Insomnia

The Insomnia Severity Index, developed by [Bastien et al. \(2001\)](#), is a seven-item scale (e.g., *I have trouble staying asleep*) with a five-point (0–4) scale ranging from “none” to “very severe”. It was used to evaluate the insomnia. The scale had good applicability in the Chinese adolescent population ([Chung et al., 2011](#)). The Guttman’s lambda coefficient in the current study was 0.92. Fit indices for the confirmatory factor analysis: $\chi^2 = 222.55, df = 14, CFI = 0.94, TLI = 0.91, RMSEA = 0.17, SRMR = 0.04$, with good scale construct validity.

2.2.3. Parent-child relationship

The Parent-child Intimacy Questionnaire was developed by [Buchanan et al. \(1991\)](#) to rate father-child and mother-child relationships separately with nine items each (e.g., *you feel close to your father/mother*) on a five-point scale ranging from 1 for “not at all” to 5 for “completely”. The Guttman’s lambda coefficient for the father-child relationship was 0.84, and that for the mother-child relationship was 0.85. The scale had good construct validity. The fit indices for the confirmatory factor analysis for the father-child relationship were: $\chi^2 = 118.94, df = 27, CFI = 0.95, TLI = 0.94, RMSEA = 0.08, SRMR = 0.04$. Fit indices for the mother-child relationship were: $\chi^2 = 109.62, df = 27, CFI = 0.95, TLI = 0.95, RMSEA = 0.07, SRMR = 0.04$. The current study used the mean of both as the parent-child relationship score, and paired sample *t*-test scores showed no significant differences ($M \pm SD$ father-child = $30.27 \pm 8.40, M \pm SD$ mother-child = $30.28 \pm 8.39, t = 0.01, p = .99$).

2.2.4. Peer relationship

Using the Short form-the Friendship Quality Questionnaire developed by [Parker and Asher \(1993\)](#), which contains eighteen items (e.g., *this friend and I make each other feel important and special*), a five-point scale is used, with 1 representing “not at all” and 5 representing “completely”. The Guttman’s lambda coefficient in this study was 0.83, and the confirmatory factor analysis showed good scale construct validity: $\chi^2 = 591.09, df = 135, CFI = 0.83, TLI = 0.81, RMSEA = 0.08, SRMR = 0.07$.

2.2.5. Teacher-student relationship

Due to the special nature of the teacher-student relationship in China, the head teacher has the greatest influence on the students, so the current study used the Teacher-Student Relationship Questionnaire ([Jiang, 2004](#)). The scale was used to assess the relationship between student and head teacher and consisted of eight items (e.g., *the head teacher is easy to get close to*), rated on a five-point scale, from 1 being “never” to 5 being “always”. The Guttman’s lambda coefficient in this study was 0.92 and the fit indices for the confirmatory factor analysis

were: $\chi^2 = 170.92, df = 20, CFI = 0.96, TLI = 0.94, RMSEA = 0.12, SRMR = 0.03$.

2.3. Analysis method

2.3.1. Data availability analysis

The data for the current study were obtained from self-reports, and the problem of common method bias that exists cannot be circumvented. The five questionnaires were explored for common method bias by Harman’s one-way method, validated factor analysis of the one-way model, and the potential error variable controllability method. Second, the Hardy-Weinberg genetic equilibrium test was performed to examine gene sampling for serotonin system polygenes.

2.4. Serotonergic multilocus genetic profile score (MGPS) calculation

The serotonin genetic profile score (MGPS) is an approach inspired by previous research ([Starr et al., 2021](#)), whereby it is computed by encoding the presence (1) or absence (0) of risk genotypes (see [Table 1](#)) for SNPs (with a heterozygous allele being assigned a code of 0.5 in cases of allele effects rather than genotype effects). The individual SNP codes are summed, and higher MGPS values reflect higher genetic risk.

2.5. Correlation analysis and test of moderating mediation model

1) correlation analysis was used to explore the correlation between the core variables; 2) three regression equations were constructed to test the moderating mediation model, eq. 1 estimates the moderating effect of MGPS on the effect of interpersonal relationship on adolescent depressive symptoms; eq. 2 estimates the effect of interpersonal relationship on insomnia; eq. 3 estimates the moderating effect of MGPS on

Table 1
The coding scheme of serotonin related snps and their associations with the serotonin function.

SNP	Associations with serotonin function	Risk allele
TPH2 rs4570625	In accordance with Vrshek-Schallhorn et al. (2015) , TPH2 rs4570625 G-allele and HTR1A rs6295 G-allele are risk alleles that affect serotonin function	G
TPH2 rs4290270	In a meta-analysis, Liu et al. (2022a, 2022b) identified that rs4290270 (T allele) may serve as a significant susceptibility factor for depression; a study revealed that individuals who have experienced non-specific trauma and possess the TT homozygous genotype for rs4290270 exhibit the highest propensity for suicide attempts (Karanović et al., 2017).	T
SCL6A4 rs1042173	In a study exploring the association between alcohol craving and alcohol use tendencies with the SLC6A4 gene, the SCL6A4 gene rs1042173 TT genotype was found to be significantly associated with higher levels of alcohol urge and alcohol craving when exposed to alcohol cues (Ait-Daoud et al., 2012).	T
HTR1A rs1364043	Takekita et al. (2015) found a significant correlation between the rs1364043 T allele and the percent change in the PANSS 5-factor negative score.	T
HTR1A rs6295	In accordance with Vrshek-Schallhorn et al. (2015) , TPH2 rs4570625 G-allele and HTR1A rs6295 G-allele are risk alleles that affect serotonin function	G
HTR2A rs7997012	The A carriers (A/A + A/G) showed significantly higher rate of suicide attempt compared to the G/G when there is a history of child abuse (Shinozaki et al., 2013)	A
HTR2A rs6311	The G allele of rs6311 is related to worse mental health and higher alexithymia as compared to the AA genotype (Li et al., 2020)	G
MAOA rs6323	Maternal supportive parenting significantly negatively predicted depression in female adolescents with the GG genotype, but its predictive effect on depression in female adolescents with the TT genotype was not significant (Cao et al., 2016).	G

the effect of interpersonal relationship and insomnia on adolescent depressive symptoms, and finally, the moderating effect of MGPS was explored using Johnson-Neyman’s moderating effect plots (Hayes and Matthes, 2009).

2.6. Sensitivity testing

First, the effect of every single polymorphism-environment interaction on adolescent depressive symptoms was examined; second, the model was revalidated by constructing eight independent seven-genotype MGPSs by removing each locus one at a time from the whole MGPS, which were used to explore whether any one locus contributed more to the gene–environment interaction than the others.

3. Results

3.1. Data availability test

Since the data is derived from self-reporting, there are often missing values. The results of the missing data analysis showed: 0 missing variable, 84 incomplete cases (16.19 %), 858 missing values (2.30 %), with a small missing rate. The Expectation-Maximisation (E-M) approach was used in the study. E-M approach assumes that the type of missing data is Missing at Random (MAR) and uses the current data information and the specified model to make a “best guess” for the missing data (Bennett, 2001).

Since the data was obtained from self-report, common method bias is a potential problem. The questionnaires were explored for common method bias by Harman’s one-way method, confirmatory factor analysis and potential error variable controllable method (Jordan and Troth, 2020). The Harman one-way method was used to conduct exploratory factor analysis on all items of the five questionnaires. The results of the principal component analysis showed that a total of 10 factors had characteristic roots >1, and the variance explained by the first factor was 18.11 % (below the critical index of 40 %). In addition, packing the three highest item loadings for each scale as the corresponding latent variable indicators, the result of the confirmatory factor analysis showed a poor model fit: $\chi^2 = 2020.23$, $df = 54$, CFI = 0.39, TLI = 0.25, RMSEA = 0.26, and SRMR = 0.20. Finally, using the potential error variable controllable method, the five-factor model fit was better before the introduction of common method bias: $\chi^2 = 63.58$, $df = 80$, CFI = 0.99, TLI = 0.99, RMSEA = 0.01, SRMR = 0.02. After the introduction of common method bias, $\chi^2 = 33.18$, $df = 36$, CFI = 0.99, TLI = 0.99, RMSEA = 0.01, SRMR = 0.01, with no significant differences between before and after. In summary, the results indicated that there was no serious common method bias.

Hardy–Weinberg equilibrium is tested in unrelated individuals using a χ^2 goodness-of-fit test that compares expected and observed numbers of heterozygotes and homozygotes (Zhou et al., 2009). The results showed in Table 2, and the genotypic observations were in better agreement with the expected number. Since MAOArs6323 is located on

the X chromosome, it was not tested. In addition, the minor allele frequency (MAF) in this study were as follows: THP2rs4570625: 0.43; THP2rs4290270: 0.44; SCL6A4rs1042173: 0.20; HTR1Ars1364043: 0.39; HTR1Ars6295: 0.24; HTR2Ars7997012: 0.26; HTR2Ars6311: 0.39; MAOArs6323: 0.58. They were all >0.05 and similar to the frequency of the minimum allele frequency MAF (1000 g-CHBS) in the Chinese Han population in Beijing from the Thousand Genomes Project: THP2rs4570625: 0.47; THP2rs4290270: 0.49; SCL6A4rs1042173: 0.21; HTR1Ars1364043: 0.39; HTR1Ars6295: 0.24; HTR2Ars7997012: 0.28; HTR2Ars6311: 0.41; MAOArs6323: 0.38, the results show that this sampling is good.

3.2. MGPS calculation and correlation analysis

Using linear regression to explore the effects of different genetic loci on follow-up depressive symptoms. The results showed the following: THP2rs4570625 ($\beta = 0.13$, $p = .003$), THP2rs4290270 ($\beta = 0.078$, $p = .07$), SCL6A4rs1042173 ($\beta = 0.01$, $p = .92$), HTR1Ars1364043 ($\beta = -0.002$, $p = .97$), HTR1Ars6295 ($\beta = 0.04$, $p = .42$), HTR2Ars7997012 ($\beta = -0.06$, $p = .14$), HTR2Ars6311 ($\beta = 0.05$, $p = .23$), and MAOArs6323 ($\beta = -0.07$, $p = .12$). Only the main effect of the THP2 gene rs4570625 polymorphism was significant, which significantly predicted depression as the mutant allele increased. Table 3 shows that the MGPS was not correlated with environmental variables (i.e., genetic and environmental indicators were independent of each other, consistent with the genetic-environmental interaction study paradigm, Dunn et al., 2011). The other core variables were significantly correlated in two waves, with a significant increase in depression in wave 3 ($t = 4.09$, $p < .001$).

3.3. The effect of interpersonal relationship on adolescent depressive symptoms: a moderating mediation test

Studies have found that only-child, gender, parental marital status and family economic level affected on adolescent depressive symptoms (Zeng et al., 2023), so they were included as covariates in the test. All continuous variables were standardized. The moderating mediation model were analyzed using the PROCESS macro for SPSS (Hayes and Scharkow, 2013). The bias-corrected 95 % confidence interval (CI) was calculated with 5000 bootstrapping resamples. Model 15 was used to examine the moderating mediation effect. The results showed (Table 4, Table 5 and Table 6) that gender, parental marital status, family economic level and interpersonal relationship predicted adolescent depressive symptoms ($|\beta s| > 0.12$, $ps < 0.02$) in eq. 1; parent-child and teacher-student relationship influenced insomnia ($\beta_{\text{parent-child}} = -0.16$, $p < .001$; $\beta_{\text{teacher-student}} = 0.10$, $p = .01$) in eq. 2; gender, parental marital status, family economic level, interpersonal relationships and MGPS significantly affected adolescent depressive symptoms ($|\beta s| > 0.08$, $ps < 0.03$), the interaction effects of insomnia and MGPS were significant ($|\beta s| > 0.09$, $ps < 0.02$), and the interaction terms of teacher-student relationship and MGPS was also borderline significant ($\beta = -0.06$, p

Table 2
Distribution of gene genotypes and Hardy-Weinberg genetic equilibrium test.

Locus				Distribution of gene genotypes				Locus				Distribution of gene genotypes			
THP2rs4570625		T/T	180 (34.3 %)	T/G	237 (45.1 %)	G/G	108 (20.6 %)	THP2rs4290270		A/A	172 (32.8 %)	A/T	249 (47.4 %)	T/T	104 (19.8 %)
χ^2	3.34			p			0.19	χ^2	0.65			p			0.72
SCL6A4rs1042173		C/C	333 (63.4 %)	C/A	171 (32.6 %)	A/A	21 (4.0 %)	THR1Ars1364043		G/G	198 (37.7 %)	G/T	247 (47.0 %)	T/T	80 (15.2 %)
χ^2	0.03			p			0.99	χ^2	0.04			p			0.98
HTR1Ars6295		G/G	292 (55.6 %)	G/C	208 (39.8 %)	C/C	25 (4.8 %)	HTR2Ars7997012		G/G	293 (55.8 %)	G/A	190 (36.2 %)	A/A	42 (8.0 %)
χ^2	2.48			p			0.29	χ^2	2.00			p			0.37
HTR2Ars6311		T/T	199 (37.9 %)	T/C	239 (45.5 %)	C/C	87 (16.6 %)	MAOArs6323		G/G	245 (46.7 %)	T/G	122 (23.2 %)	T/T	158 (30.1 %)
χ^2	1.12			p			0.57	–							

Table 3

Means and standard deviations of the major variables along with their correlations.

Variables	1	2	3	4	5	6	7
MGPS	1						
Parent-child	−0.07	1					
Peer	−0.03	0.27**	1				
Teacher-student	−0.02	0.27**	0.15**	1			
Insomnia	0.02	−0.17**	−0.06	−0.11*	1		
Baseline depressive symptoms	0.04	−0.42**	−0.24**	−0.22**	0.34**	1	
Follow-up depression symptoms	0.10**	−0.27**	−0.17**	−0.20**	0.38**	0.45**	1
<i>M</i> ± <i>SD</i>	6.89 ± 1.84	31.36 ± 8.19	66.25 ± 12.26	35.14 ± 7.29	5.27 ± 6.64	7.37 ± 5.65	8.50 ± 6.38

Note: 1 is MGPS; 2 is parent-child relationship; 3 is peer relationship; 4 is teacher-student relationship; 5 is insomnia; 6 is baseline depressive symptoms; 7 is follow-up depressive symptoms; **p* < .05, ***p* < .01, ****p* < .001, same below.

Table 4

Effect of parent-child relationship on adolescent depressive symptoms.

Variables	Eq. (1) (Dependent variable: Y)				Eq. (2) (Dependent variable:M)				Eq. (3) (Dependent variable:Y)			
	β	<i>t</i>	<i>p</i>	95 % CI	β	<i>t</i>	<i>p</i>	95 % CI	β	<i>t</i>	<i>p</i>	95 % CI
Gender	0.22	2.59**	0.01	[0.05, 0.38]	0.15	1.78	0.08	[−0.02, 0.32]	0.17	2.23*	0.03	[0.02, 0.33]
Only-child	−0.08	−0.92	0.36	[−0.25, 0.09]	0.01	0.10	0.92	[−0.16, 0.18]	−0.08	−1.07	0.29	[−0.24, 0.07]
Marital Status	0.12	2.52*	0.01	[0.01, 0.23]	0.08	1.66	0.10	[−0.02, 0.18]	0.09	1.94	0.05	[−0.001, 0.18]
Economic level	0.15	2.58*	0.01	[0.03, 0.28]	0.11	1.77	0.08	[−0.02, 0.18]	0.12	2.12*	0.03	[0.01, 0.22]
X	−0.27	−6.40***	<0.001	[−0.35, −0.19]	−0.16	−3.76***	<0.001	[−0.24, −0.08]	−0.21	−5.40***	<0.001	[−0.29, −0.14]
M									0.33	8.32***	<0.001	[0.25, 0.41]
W									0.09	2.38*	0.02	[0.02, 0.17]
X*W									−0.05	−1.22	0.22	[−0.14, 0.03]
M*W									0.10	2.44*	0.02	[0.02, 0.18]
R ²	0.11				0.05				0.23			
F	12.92***				4.98***				17.45***			

Note: Gender, 1 for male, 2 for female,

Only-child, 1 for yes, 2 for no.

Parents' marital status, 1 for first married, 2 for divorced, 3 for remarried, 4 for other.

Family economic level, 1 for very good, 2 for good, 3 for average, 4 for poor, 5 for very poor.

X is different interpersonal relationship, M is insomnia, W is MGPS, Y is adolescent depressive symptoms.

Bolded font indicates significant, same below.

Table 5

Effect of peer relationship on adolescent depressive symptoms.

Variables	Eq. (1) (Dependent variable: Y)				Eq. (2) (Dependent variable:M)				Eq. (3) (Dependent variable:Y)			
	β	<i>t</i>	<i>p</i>	95 % CI	β	<i>t</i>	<i>p</i>	95 % CI	β	<i>t</i>	<i>p</i>	95 % CI
Gender	0.26	2.97**	0.003	[0.09, 0.42]	0.16	1.88	0.06	[−0.01, 0.34]	0.20	2.54*	0.01	[0.05, 0.36]
Only-child	−0.06	−0.64	0.52	[−0.22, 0.11]	0.03	0.31	0.76	[−0.14, 0.20]	−0.07	−0.88	0.38	[−0.23, 0.09]
Marital status	0.12	2.48*	0.01	[0.02, 0.24]	0.09	1.77	0.08	[−0.01, 0.19]	0.09	1.87	0.06	[−0.004, 0.18]
Economic level	0.14	2.39*	0.02	[0.01, 0.27]	0.11	1.76	0.08	[−0.01, 0.23]	0.10	1.82	0.07	[−0.01, 0.21]
X	−0.17	−3.94***	<0.001	[−0.25, −0.08]	−0.05	−1.15	0.25	[−0.14, 0.04]	−0.15	−3.72***	<0.001	[−0.23, −0.07]
M									0.36	8.92***	<0.001	[0.28, 0.43]
W									0.09	2.28*	0.02	[0.01, 0.17]
X * W									−0.02	−0.57	0.57	[−0.11, 0.06]
M * W									0.10	2.38*	0.02	[0.02, 0.17]
R ²	0.07				0.02				0.21			
F	7.63***				2.36*				15.06***			

= .09) in eq. 3. The above results suggest that interpersonal relationship, insomnia, MGPS and adolescent depressive symptoms constitute a moderating mediation model.

As tests of robustness, following recommendations of Dick et al. (2015) and Keller (2014), we also conducted models in which covariates were entered as main and interaction effects (resulting in twenty additional covariates terms, e.g., gender × G, gender × E.). In these models, the G*E interaction remained significant ($\Delta R^2 = 0.012$, *p* = .003 for parent-child model; $\Delta R^2 = 0.015$, *p* = .001 for peer model; $\Delta R^2 = 0.013$, *p* = .002 for teacher-student model).

Fig. 2. provide of parent-child relationship (top left), peer relationship (top right), and insomnia scores' lower and upper boundaries of the zone of significance using the Johnson-Neyman approach in PROCESS. It showed that when the standard deviation was greater than −1.78

(parent-child model) /−1.97 (peer model) /−1.93 (teacher-student model), as the MGPS increased, the slopes of insomnia on adolescent depressive symptoms increased and the effects were enhanced; further when the MGPS standard deviation was greater than −1.36, the effects of teacher-student relationship on depressive symptoms showed a similar situation with increasing slope and stronger prediction as the score increased.

3.4. Sensitivity tests

3.4.1. Effects of single locus-environment interactions on adolescent depressive symptoms

First, regression analysis was performed to explore the interactions of any locus with the environment on depressive symptoms. The results

Table 6
Effect of teacher-student relationship on adolescent depressive symptoms.

Variables	Eq. (1) (Dependent variable: Y)				Eq. (2) (Dependent variable:M)				Eq. (3) (Dependent variable:Y)			
	β	t	p	95 % CI	β	t	p	95 % CI	β	t	p	95 % CI
Gender	0.21	2.42**	0.02	[0.04, 0.37]	0.15	1.71	0.09	[−0.02, 0.32]	0.15	1.89	0.06	[−0.01, 0.31]
Only-child	−0.04	−0.49	0.63	[−0.20, 0.13]	0.03	0.35	0.72	[−0.14, 0.20]	−0.05	−0.69	0.50	[−0.21, 0.10]
Marital status	0.15	2.94*	0.003	[0.04, 0.25]	0.10	1.92	0.06	[−0.002, 0.20]	0.11	2.33*	0.02	[0.02, 0.20]
Economic level	0.15	2.45*	0.02	[0.02, 0.28]	0.10	1.72	0.09	[−0.01, 0.22]	0.10	1.89	0.06	[−0.004, 0.21]
X	−0.19	−4.67***	<0.001	[−0.29, −0.11]	−0.10	−2.44***	0.01	[−0.19, −0.02]	−0.18	−4.36***	<0.001	[−0.26, −0.10]
M									0.35	8.67***	<0.001	[0.27, 0.42]
W									0.08	2.13*	0.03	[0.01, 0.16]
X * W									−0.06	−1.66	0.09	[−0.12, 0.01]
M * W									0.09	2.33*	0.02	[0.01, 0.17]
R ²	0.08				0.03				0.22			
F	8.93***				3.30**				15.78***			

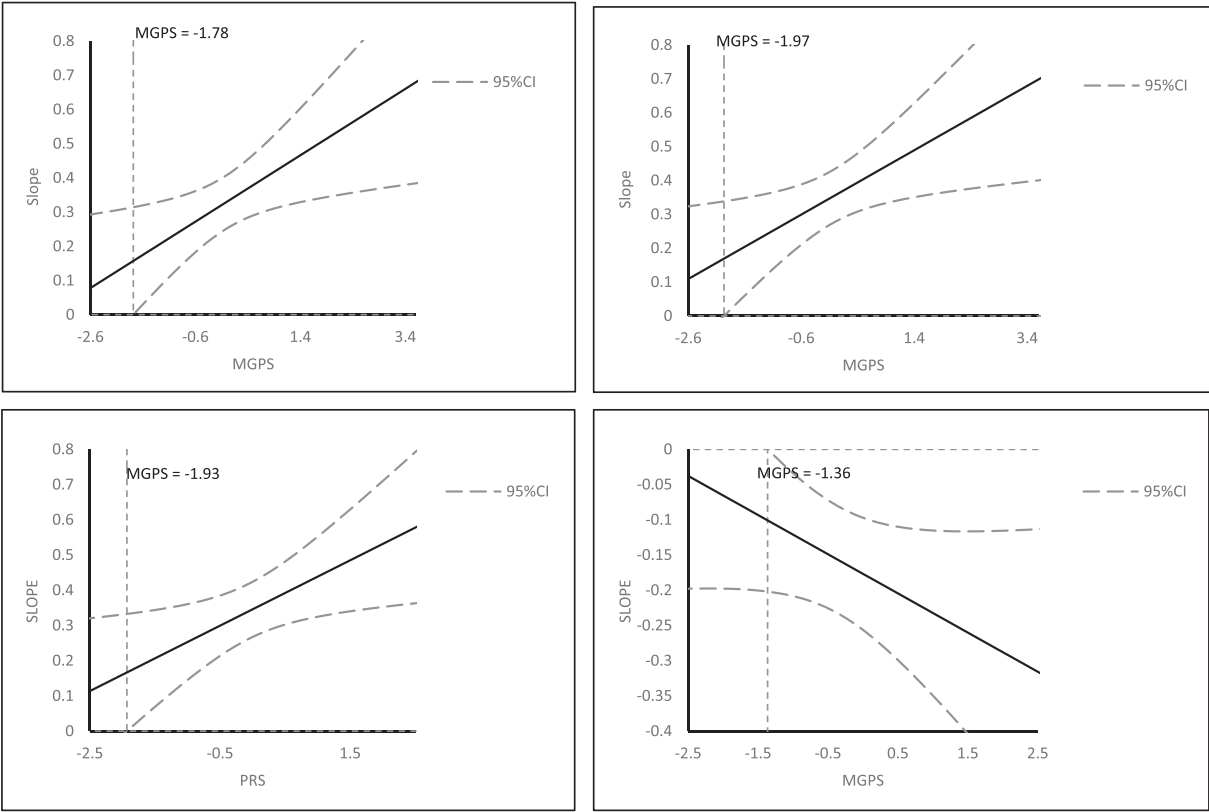


Fig. 2. Johnson-Neyman plots of the moderating role of MGPS in the effects of and insomnia (top and bottom left) and teacher-student relationship on depressive symptoms.

showed that only the THP2 gene rs4570625 with the environment ($\beta_{\text{parent-child*rs4570625}} = -0.13, p = .02$; $\beta_{\text{peer*rs4570625}} = -0.12, p = .03$; $\beta_{\text{insomnia*rs4570625}} = 0.15/0.14/0.15, p = .01$) and the interaction of THP2 gene rs4290270 with teacher-student relationship ($\beta_{\text{teacher-student*rs4290270}} = -0.13, p = .02$) affected on adolescent depressive symptoms. No other interaction effects of any individual locus with the environment were significant ($|\beta| < 0.11, p > 0.06$). The results suggest that MGPS are more advantageous in terms of genetic explanatory rates of depressive symptoms compared to single gene studies.

3.4.2. Effects of independent seven-genotype MGPSs-environment interactions on adolescent depressive symptoms

In addition, eight independent seven-genotype MGPSs were constructed by removing each locus one at a time to explore their interaction effects of them with the environment on adolescent depressive symptoms. The results showed that in the parent-child model, only after

taking out the THP2 gene rs4570625 ($\beta_{\text{parent-child*rs4570625}} = -0.01, p = .86$; $\beta_{\text{insomnia*rs4570625}} = 0.05, p = .08$) and HTR2A gene rs6311 ($\beta_{\text{parent-child*rs6311}} = -0.02, p = .38$; $\beta_{\text{insomnia*rs6311}} = 0.04, p = .06$), the interaction effects of MGPSs with the environment appeared significantly altered; In the peer model, only after taking out the THP2 gene rs4570625 ($\beta_{\text{peer*rs4570625}} = 0.02, p = .54$; $\beta_{\text{insomnia*rs4570625}} = 0.03, p = .08$) and HTR2A gene rs6311 ($\beta_{\text{peer*rs6311}} = -0.01, p = .64$; $\beta_{\text{insomnia*rs6311}} = 0.04, p = .08$), the interaction effects of MGPSs with the environment appeared significantly altered; In the teacher-student model, only after taking out the THP2 gene rs4570625 ($\beta_{\text{teacher-student*rs4570625}} = -0.04, p = .08$; $\beta_{\text{insomnia*rs4570625}} = 0.04, p = .14$), the interaction effects of MGPSs with the environment appeared significantly altered.

4. Discussion

The influence of gene-environment interactions on individual

development has been a hot area in psychology, psychiatry and molecular genetics. Based on the polygenic and environmental research paradigm of depression, the current study explored the effect of interpersonal relationship on adolescent depressive symptoms, as well as the mediating role of insomnia and the moderating role of the serotonergic multilocus genetic profile score. The study adds evidence to the theory of depression development and has some implications for intervention.

Ecological systems theory states that the interaction between the individual and the external ecosystem influences the physical and mental development (Bronfenbrenner, 1979). The family and school, as the innermost microsystem of the ecosystem, are the environments where adolescents engage in the most social activities and have the greatest influence (Platt et al., 2013). Parent-child, peer, and teacher-student relationships, the most important components of the innermost microsystem, have been shown in previous research to improve adolescents' emotional support and reduce loneliness, anxiety, and depression (Clayborne et al., 2019).

The current results showed that interpersonal relationship significantly affected the development of adolescent depressive symptoms, consistent with previous research. In a study exploring parenting styles and psychiatric disorders in 6483 US adolescents, the results showed that high parental control was associated with high levels of depression, anxiety, eating and behavior disorders, while a high level of parental care reduced adolescent social phobia and alcohol use (Eun et al., 2018). In addition, good peer relationship is an important source of belonging and self-esteem for adolescents and a protective factor against adolescent depression (Baumeister and Leary, 1995). Conversely, negative peer relationship (such as, rejection, exclusion, etc.) could be a risk factor for adolescent depression by increasing adolescents' feelings of exclusion (Platt et al., 2013). In their study of teacher-student relationships and behavioral problems in elementary school students, O'Connor et al. (2011) found that high-quality teacher-student relationship influenced externalizing problems and could serve as protective factors to help prevent long-term internalizing problem trajectories.

On the other hand, related psychological counseling and intervention cases have found that adolescents are often prone to sleep problems including insomnia and excessive dreaming when facing interpersonal problems or academic stress, which leads to psychological problems such as depression and anxiety. In a study exploring the effect of cognitive decline on depression in older adults, results also found a mediating role of insomnia (Liu et al., 2022a, 2022b). Given that both interpersonal functioning and insomnia independently were predictors of treatment outcome for depression symptoms, it seems likely that addressing insomnia in the context of interpersonal relationship may lead to more robust treatment outcomes for adolescent depressive symptoms. A greater focus on interpersonal relationship and its underlying mechanisms in the context of treatments for depression will be an important area for future research, particularly among adolescents where insomnia is prevalent and the mechanisms of its effect on depression are complex. This study found that interpersonal relationship influenced adolescent depressive symptoms through insomnia, providing a new idea for the mechanism of depression development.

Transactional development theory proposes that the individual's behavior is the result of the interaction of innate factors and the acquired environment during a specific developmental period. Unlike previous single polymorphism-environment studies, this study selected the serotonin system genes that have important effects on depression, including all processes of serotonin synthesis, transport, receptor and hydrolysis. The findings showed a significant main effect of the THP2 gene rs4570625 polymorphism. In a systematic review and meta-analysis that included 166 studies, Ottenhof et al. (2018) found that the number of THP2 gene rs4570625 polymorphism G alleles was associated with psychiatric disorders such as mood disorders, monophasic depression, anxiety and psychotic disorders. The above results suggest that the THP2 gene rs4570625 polymorphic G allele is a key risk factor for depression.

Moreover, the results of moderating mediation effects showed a

significant interaction. Specifically, as the MPGS increased, the effects of insomnia on adolescent depressive symptoms increased; further, when the MPGS increased, the effect of teacher-student relationship on depression showed a similar phenomenon. Different from the diathesis-stress model, which states that the combination of 'vulnerability genetic' and environmental stress leads to depression, the differential susceptibility model proposes that genetics is not 'vulnerability' but 'plasticity', namely, individuals carrying genetic factors of "plasticity" have a higher risk of depression due to the influence of the negative environment. However, they are also affected by positive environments and have better performance (Bakermans-Kranenburg and Van Ijzendoorn, 2011). Previous studies have tended to select negative environments, such as stressful life events and childhood abuse, to explore the interaction between genetics and environment, thus neglecting the influence of positive environments (Starr et al., 2019). The current study used two opposing factors (interpersonal relationship and insomnia). The findings support the differential susceptibility model of depression. Finally, the results of sensitivity analysis showed that the interaction between multilocus genetic and the environment had stronger explanatory power and better stability and reliability for depression than single polymorphism studies.

Based on polygenic research paradigm, this study selected key genetic (serotonergic multilocus genetic) and environmental variables (interpersonal relationship) and used a follow-up data with appropriate sample sizes to explore the mechanisms of depressive symptoms and provide empirical support for related theories. However, it should be noted that the participants were students from the same school, so the sample may not be representative enough. The tracking time was not uniform due to the pandemic, so we hope to select more representative samples in future studies. Second, some studies have found that MPGS is not limited to examining disease risk but can also be used to examine behavioral performance, brain activity and molecular measures related to environmental responses (Bulik-Sullivan et al., 2015). Third, although the scales used in this study were widely used and had good reliability and validity, the indices of the confirmatory factor analysis were not good. It is possible that the pandemic has caused students to resist the psychological questionnaire scales. Finally, although the results support the differential susceptibility model, the current approach of using regression weights might lead to effect loss if only main effects are considered.

5. Conclusion

In summary, the current study found that (1) interpersonal relationship predicted adolescent depressive symptoms; (2) insomnia mediated the effect of interpersonal relationships on adolescent depressive symptoms; (3) the THP2 gene rs4570625 polymorphism G allele was a key risk factor for depressive symptom, and the MGPS moderated the effects of teacher-student relationship and insomnia on adolescent depressive symptom. Specifically, as the MGPS increased, the effects of insomnia on adolescent depressive symptom were enhanced; further, when the MGPS score increased, the effect of teacher-student relationship on depression showed a similar phenomenon with an increased slope and enhanced prediction; and (4) the results of sensitivity analysis showed that multilocus genetic interaction with the environment had a better explanatory power and stability for depression than single polymorphism studies.

Ethical

Approval All procedures performed in studies involving human participants were in accordance with the ethical standards of the institutional and/or national research committee and with the 1964 Helsinki declaration and its later amendments or comparable ethical standards.

Informed consent

Informed consent was obtained from all individual participants included in the study.

Author contributions

Zihao Zeng participated in interpretation of the data, and drafted the manuscript; Liyi Peng and Shuangjin Liu participated in its design, performed the statistical analyses; Qin Yang, Hongcai Wang and Zhen He helped participated in the interpretation of the data, and helped draft the manuscript; Yiqiu Hu conceived of the study, provided reviews and funds. All authors read and approved the final manuscript.

Sources of funding

This work was supported by the Hunan Education Science Key Funding Project (grant numbers XJK23AJD023) and Postgraduate Scientific Research Innovation Project of Hunan Province (CX20210425). The funders had no role in study design; the data collection; writing of the manuscript; or the decision to submit the article for publication.

Declaration of competing interest

The authors declare that they have no conflict of interest.

Acknowledgements

The authors would like to thank Dr. Liu Yong for his assistance with the blood sample extraction. Thanks also to Xiuru Liu, Xingxing Yao and Zhen He for their assistance in data acquisition.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jad.2023.08.085>.

References

- Ahrberg, K., Dresler, M., Niedermaier, S., Steiger, A., Genzel, L., 2012. The interaction between sleep quality and academic performance. *J. Psychiatr. Res.* 46 (12), 1618–1622.
- Ait-Daoud, N., Seneviratne, C., Smith, J.B., Roache, J.D., Dawes, M.A., Liu, L., Johnson, B.A., 2012. Preliminary evidence for cue-induced alcohol craving modulated by serotonin transporter gene polymorphism rs1042173. *Front. Psychiatry* 3, 6.
- Andresen, E.M., Malmgren, J.A., Carter, W.B., Patrick, D.L., 1994. Screening for depression in well older adults: evaluation of a short form of the CES-D. *Am. J. Prev. Med.* 10 (2), 77–84.
- Andrews, P.W., Bharwani, A., Lee, K.R., Fox, M., Thomson Jr., J.A., 2015. Is serotonin an upper or a downer? The evolution of the serotonergic system and its role in depression and the antidepressant response. *Neurosci. Biobehav. Rev.* 51, 164–188.
- Bakermans-Kranenburg, M.J., Van Ijzendoorn, M.H., 2011. Differential susceptibility to rearing environment depending on dopamine-related genes: new evidence and a meta-analysis. *Dev. Psychopathol.* 23 (1), 39–52.
- Barton, D.A., Esler, M.D., Dawood, T., Lambert, E.A., Haikerwal, D., Brenchley, C., Lambert, G.W., 2008. Elevated brain serotonin turnover in patients with depression: effect of genotype and therapy. *Arch. Gen. Psychiatry* 65 (1), 38–46.
- Bastien, C.H., Vallières, A., Morin, C.M., 2001. Validation of the insomnia severity index as an outcome measure for insomnia research. *Sleep Med.* 2 (4), 297–307.
- Baumeister, R.F., Leary, M.R., 1995. The need to belong: desire for interpersonal attachments as a fundamental human motivation. *Psychol. Bull.* 117 (3), 497.
- Bennett, D.A., 2001. How can I deal with missing data in my study? *Aust. N. Z. J. Public Health* 25 (5), 464–469.
- Bronfenbrenner, U., 1979. *The Ecology of Human Development: Experiments by Nature and Design*. Harvard university press.
- Buchanan, C.M., Maccoby, E.E., Dornbusch, S.M., 1991. Caught between parents: adolescents' experience in divorced homes. *Child Dev.* 62 (5), 1008–1029.
- Bulik-Sullivan, B., Finucane, H.K., Anttila, V., Gusev, A., Day, F.R., Loh, P.R., Neale, B. M., 2015. An atlas of genetic correlations across human diseases and traits. *Nat. Genet.* 47 (11), 1236–1241.
- Cao, C., Wang, M., Ji, L., Wei, X., Cao, Y., Zhang, W., 2016. The MAOA rs6323 polymorphism interacts with maternal supportive parenting in predicting adolescent depression: testing the diathesis-stress and differential susceptibility hypotheses. *Acta Psychol. Sin.* 48 (1), 22.
- Chung, K.F., Kan, K.K.K., Yeung, W.F., 2011. Assessing insomnia in adolescents: comparison of insomnia severity index, Athens insomnia scale and sleep quality index. *Sleep Med.* 12 (5), 463–470.
- Clayborne, Z.M., Varin, M., Colman, I., 2019. Systematic review and meta-analysis: adolescent depression and long-term psychosocial outcomes. *J. Am. Acad. Child Adolesc. Psychiatry* 58 (1), 72–79.
- Dick, D.M., Agrawal, A., Keller, M.C., Adkins, A., Aliev, F., Monroe, S., Sher, K.J., 2015. Candidate gene–environment interaction research: reflections and recommendations. *Perspect. Psychol. Sci.* 10 (1), 37–59.
- Dong, Y., Wang, H., Luan, F., Li, Z., Cheng, L., 2021. How children feel matters: teacher–student relationship as an indirect role between interpersonal trust and social adjustment. *Front. Psychol.* 11, 581235.
- Dunn, E.C., Uddin, M., Subramanian, S.V., Smoller, J.W., Galea, S., Koenen, K.C., 2011. Research review: gene–environment interaction research in youth depression – a systematic review with recommendations for future research. *J. Child Psychol. Psychiatry* 52 (12), 1223–1238.
- Eun, J.D., Paksarian, D., He, J.P., Merikangas, K.R., 2018. Parenting style and mental disorders in a nationally representative sample of US adolescents. *Soc. Psychiatry Psychiatr. Epidemiol.* 53, 11–20.
- Fabbri, C., Marsano, A., Serretti, A., 2013. Genetics of serotonin receptors and depression: state of the art. *Curr. Drug Targets* 14 (5), 531–548.
- Gong, L., He, C., Yin, Y., Wang, H., Ye, Q., Bai, F., Xie, C., 2017. Mediating role of the reward network in the relationship between the dopamine multilocus genetic profile and depression. *Front. Mol. Neurosci.* 10, 292.
- Gordon, A.M., Carrillo, B., Barnes, C.M., 2021. Sleep and social relationships in healthy populations: a systematic review. *Sleep Med. Rev.* 57, 101428.
- Gorostiaga, A., Aliri, J., Balluerka, N., Lameirinhas, J., 2019. Parenting styles and internalizing symptoms in adolescence: a systematic literature review. *Int. J. Environ. Res. Public Health* 16 (17), 3192.
- Gunn, H.E., Troxel, W.M., Hall, M.H., Buysse, D.J., 2014. Interpersonal distress is associated with sleep and arousal in insomnia and good sleepers. *J. Psychosom. Res.* 76 (3), 242–248.
- Hammen, C., 2018. Risk factors for depression: an autobiographical review. *Annu. Rev. Clin. Psychol.* 14, 1–28.
- Hayes, A.F., Matthes, J., 2009. Computational procedures for probing interactions in OLS and logistic regression: SPSS and SAS implementations. *Behav. Res. Methods* 41 (3), 924–936.
- Hayes, A.F., Scharkow, M., 2013. The relative trustworthiness of inferential tests of the indirect effect in statistical mediation analysis: does method really matter? *Psychol. Sci.* 24 (10), 1918–1927.
- Hosker, D.K., Elkins, R.M., Potter, M.P., 2019. Promoting mental health and wellness in youth through physical activity, nutrition, and sleep. *Child Adolesc. Psychiatr. Clin.* 28 (2), 171–193.
- Hotamisligil, G.S., Breakefield, X.O., 1991. Human monoamine oxidase a gene determines levels of enzyme activity. *Am. J. Hum. Genet.* 49 (2), 383.
- Hou, Z., Jiang, W., Li, F., Liu, X., Hou, Z., Yin, Y., Yuan, Y., 2023. Linking individual variability in functional brain connectivity to polygenic risk in major depressive disorder. *J. Affect. Disord.* 329, 55–63.
- Hu, Y., Zeng, Z., Peng, L., Zhan, L., Liu, S., Ouyang, X., Li, Z., 2022. The effect of childhood maltreatment on college students' depression symptoms: the mediating role of subjective well-being and the moderating role of MAOA gene rs6323 polymorphism. *Eur. J. Dev. Psychol.* 19 (3), 438–457.
- Jansson, M., McCarthy, S., Sullivan, P.F., Dickman, P., Andersson, B., Orelund, L., Pedersen, N.L., 2005. MAOA haplotypes associated with thrombocyte-MAO activity. *BMC Genet.* 6 (1), 1–9.
- Januar, V., Saffery, R., Ryan, J., 2015. Epigenetics and depressive disorders: a review of current progress and future directions. *Int. J. Epidemiol.* 44 (4), 1364–1387.
- Jiang, G.R., 2004. Class environment in the Chinese school system: structure and measurement. *Psychol. Sci.* 27 (4), 839–843.
- Johnco, C., Rapee, R.M., 2018. Depression literacy and stigma influence how parents perceive and respond to adolescent depressive symptoms. *J. Affect. Disord.* 241, 599–607.
- Jordan, P.J., Troth, A.C., 2020. Common method bias in applied settings: the dilemma of researching in organizations. *Aust. J. Manag.* 45 (1), 3–14.
- Karanović, J., Ivković, M., Jovanović, V.M., Šviković, S., Pantović-Stefanović, M., Brkušanić, M., Savić-Pavićević, D., 2017. Effect of childhood general traumas on suicide attempt depends on TPH2 and ADAR1 variants in psychiatric patients. *J. Neural Transm.* 124, 621–629.
- Keller, M.C., 2014. Gene × environment interaction studies have not properly controlled for potential confounders: the problem and the (simple) solution. *Biol. Psychiatry* 75 (1), 18–24.
- Larson, C.L., Taubitz, L.E., Robinson, J.S., 2010. MAOA T941G polymorphism and the time course of emotional recovery following unpleasant pictures. *Psychophysiology* 47 (5), 857–862.
- Li, L., Wang, Y.Y., Wang, S.B., Zhang, L., Li, L., Xu, D.D., Xiang, Y.T., 2018. Prevalence of sleep disturbances in Chinese university students: a comprehensive meta-analysis. *J. Sleep Res.* 27 (3), e12648.
- Li, X., He, L., Liu, J., Guo, W., Wang, Q., Fang, P., Gong, P., 2020. The rs6311 of serotonin receptor 2A (5-HT2A) gene is associated with alexithymia and mental health. *J. Affect. Disord.* 272, 277–282.
- Liu, Z.L., Wang, X.Q., Liu, M.F., Ye, B.J., 2022a. Meta-analysis of association between TPH2 single nucleotide polymorphism and depression. *Neurosci. Biobehav. Rev.* 134, 104517.

- Liu, X., Xia, X., Hu, F., Hao, Q., Hou, L., Sun, X., Dong, B., 2022b. The mediation role of sleep quality in the relationship between cognitive decline and depression. *BMC Geriatr.* 22 (1), 1–12.
- Longobardi, C., Settanni, M., Lin, S., Fabris, M.A., 2021. Student–teacher relationship quality and prosocial behaviour: the mediating role of academic achievement and a positive attitude towards school. *Br. J. Educ. Psychol.* 91 (2), 547–562.
- Magson, N.R., Freeman, J.Y., Rapee, R.M., Richardson, C.E., Oar, E.L., Fardouly, J., 2021. Risk and protective factors for prospective changes in adolescent mental health during the COVID-19 pandemic. *J. Youth Adolesc.* 50 (1), 44–57.
- Malhi, G.S., Mann, J.J., 2018. Depression. *Lancet* 392 (10161), 2299–2312.
- Mandelli, L., Antypa, N., Nearchou, F.A., Vaipopoulos, C., Stefanis, C.N., Serretti, A., Stefanis, N.C., 2012. The role of serotonergic genes and environmental stress on the development of depressive symptoms and neuroticism. *J. Affect. Disord.* 142 (1–3), 82–89.
- McKetta, S., Keyes, K.M., 2019. Oral contraceptive use and depression among adolescents. *Ann. Epidemiol.* 29, 46–51.
- Nonen, S., Kato, M., Takekita, Y., Wakeno, M., Sakai, S., Serretti, A., Kinoshita, T., 2016. Polymorphism of rs3813034 in serotonin transporter gene SLC6A4 is associated with the selective serotonin and serotonin-norepinephrine reuptake inhibitor response in depressive disorder: sequencing analysis of SLC6A4. *J. Clin. Psychopharmacol.* 36 (1), 27–31.
- O'Connor, E.E., Dearing, E., Collins, B.A., 2011. Teacher-child relationship and behavior problem trajectories in elementary school. *Am. Educ. Res. J.* 48 (1), 120–162.
- Ottenhof, K.W., Sild, M., Lévesque, M.L., Ruhé, H.G., Booij, L., 2018. TPH2 polymorphisms across the spectrum of psychiatric morbidity: a systematic review and meta-analysis. *Neurosci. Biobehav. Rev.* 92, 29–42.
- Palmer, C.A., Alfano, C.A., 2017. Sleep and emotion regulation: an organizing, integrative review. *Sleep Med. Rev.* 31, 6–16.
- Pandi-Perumal, S.R., Monti, J.M., Burman, D., Karthikeyan, R., BaHammam, A.S., Spence, D.W., Narashimhan, M., 2020. Clarifying the role of sleep in depression: a narrative review. *Psychiatry Res.* 291, 113239.
- Parker, J.G., Asher, S.R., 1993. Friendship and friendship quality in middle childhood: links with peer group acceptance and feelings of loneliness and social dissatisfaction. *Dev. Psychol.* 29 (4), 611.
- Platt, B., Kadosh, K.C., Lau, J.Y., 2013. The role of peer rejection in adolescent depression. *Depress. Anxiety* 30 (9), 809–821.
- Resnick, B., Klinedinst, N.J., Yerges-Armstrong, L., Choi, E.Y., Dorsey, S.G., 2015. The impact of genetics on physical resilience and successful aging. *J. Aging Health* 27 (6), 1084–1104.
- Rybakowski, J.K., 2012. Bipolarity and inadequate response to antidepressant drugs: clinical and psychopharmacological perspective. *J. Affect. Disord.* 136 (2), 13–19.
- Shinozaki, G., Romanowicz, M., Mrazek, D.A., Kung, S., 2013. HTR2A gene–child abuse interaction and association with a history of suicide attempt among Caucasian depressed psychiatric inpatients. *J. Affect. Disord.* 150 (3), 1200–1203.
- Sijtsma, K., 2009. On the use, the misuse, and the very limited usefulness of Cronbach's alpha. *Psychometrika* 74 (1), 107–120.
- Stapinski, L.A., Araya, R., Heron, J., Montgomery, A.A., Stallard, P., 2015. Peer victimization during adolescence: concurrent and prospective impact on symptoms of depression and anxiety. *Anxiety Stress Coping* 28 (1), 105–120.
- Starr, L.R., Huang, M., 2019. HPA-axis multilocus genetic variation moderates associations between environmental stress and depressive symptoms among adolescents. *Dev. Psychopathol.* 31 (4), 1339–1352.
- Starr, L.R., Hammen, C., Conway, C.C., Raposa, E., Brennan, P.A., 2014. Sensitizing effect of early adversity on depressive reactions to later proximal stress: moderation by polymorphisms in serotonin transporter and corticotropin releasing hormone receptor genes in a 20-year longitudinal study. *Dev. Psychopathol.* 26 (4pt2), 1241–1254.
- Starr, L.R., Vrshek-Schallhorn, S., Stroud, C.B., 2019. Serotonergic multilocus genetic variation moderates the association between major interpersonal stress and adolescent depressive symptoms: replication and candidate environment specification. *J. Psychiatr. Res.* 117, 55–61.
- Starr, L.R., Stroud, C.B., Shaw, Z.A., Vrshek-Schallhorn, S., 2021. Stress sensitization to depression following childhood adversity: moderation by HPA axis and serotonergic multilocus profile scores. *Dev. Psychopathol.* 33 (4), 1264–1278.
- Stewart, J.G., Harkness, K.L., 2017. Testing a revised interpersonal theory of depression using a laboratory measure of excessive reassurance seeking. *J. Clin. Psychol.* 73 (3), 331–348.
- Takekita, Y., Fabbri, C., Kato, M., Nonen, S., Sakai, S., Sunada, N., Serretti, A., 2015. HTR1A gene polymorphisms and 5-HT1A receptor partial agonist antipsychotics efficacy in schizophrenia. *J. Clin. Psychopharmacol.* 35 (3), 220–227.
- Tao, S., Chattun, M.R., Yan, R., Geng, J., Zhu, R., Shao, J., Yao, Z., 2018. TPH-2 gene polymorphism in major depressive disorder patients with early-wakening symptom. *Front. Neurosci.* 12, 827.
- Tempesta, D., Socci, V., De Gennaro, L., Ferrara, M., 2018. Sleep and emotional processing. *Sleep Med. Rev.* 40, 183–195.
- Van Loon, A.W., Creemers, H.E., Vogelaar, S., Saab, N., Miers, A.C., Westenberg, P.M., Asscher, J.J., 2019. The effectiveness of school-based skills-training programs promoting mental health in adolescents: a study protocol for a randomized controlled study. *BMC Public Health* 19 (1), 1–12.
- Vrshek-Schallhorn, S., Stroud, C.B., Mineka, S., Zinbarg, R.E., Adam, E.K., Redei, E.E., Craske, M.G., 2015. Additive genetic risk from five serotonin system polymorphisms interacts with interpersonal stress to predict depression. *J. Abnorm. Psychol.* 124 (4), 776.
- Vrshek-Schallhorn, S., Stroud, C.B., Doane, L.D., Mineka, S., Zinbarg, R.E., Redei, E.E., Adam, E.K., 2019. Cortisol awakening response and additive serotonergic genetic risk interactively predict depression in two samples: the 2019 Donald F. Klein Early Career Investigator Award Paper. *Depress. Anxiety* 36 (6), 480–489.
- Vrshek-Schallhorn, S., Corneau, G.M., Grillo, A.R., Saparam, V.R., Plieger, T., Reuter, M., 2022. Additive serotonergic genetic sensitivity and cortisol reactivity to lab-based social evaluative stress: influence of severity across two samples. *Psychoneuroendocrinology* 142, 105767.
- Wang, Meiping, Zhang, Wenxin, Chen, Xinyin, 2015. The interaction between rs6295 polymorphism in the 5-HT1A gene and parenting behavior on early adolescents' depression: the verification of differential susceptibility model. *Acta Psychol. Sin.* 47 (5), 600–610.
- Weersing, V.R., Jeffreys, M., Do, M.C.T., Schwartz, K.T., Bolano, C., 2017. Evidence base update of psychosocial treatments for child and adolescent depression. *J. Clin. Child Adolesc. Psychol.* 46 (1), 11–43.
- Yang, W., Xiong, G., Garrido, L.E., Zhang, J.X., Wang, M.C., Wang, C., 2018. Factor structure and criterion validity across the full scale and ten short forms of the CES-D among Chinese adolescents. *Psychol. Assess.* 30 (9), 1186.
- Zeng, Z., Hu, Y., Peng, L., Zhan, L., Liu, S., Ouyang, X., Li, Z., 2023. The effect of negative life events on college students' depression: the mediating role of internet addiction and the moderating role of 5-HT1A gene rs6449693 polymorphism. *Curr. Psychol.* 42 (8), 6555–6563.
- Zhang, G., Stackman Jr., R.W., 2015. The role of serotonin 5-HT2A receptors in memory and cognition. *Front. Pharmacol.* 6, 225.
- Zhang, R., Bi, Y., Niu, W., Huang, X., Chen, S., Li, X., He, G., 2016. Association study of 5-HT1A, 5-HT2A polymorphisms with schizophrenia and major depressive disorder in the Han Chinese population. *Neurosci. Lett.* 635, 39–43.
- Zhang, X., Gao, F., Kang, Z., Zhou, H., Zhang, J., Li, J., Liu, B., 2022. Perceived academic stress and depression: the mediation role of mobile phone addiction and sleep quality. *Front. Public Health* 10, 760387.
- Zhou, J.J., Lange, K., Papp, J.C., Sinsheimer, J.S., 2009. A heterozygote–homozygote test of Hardy–Weinberg equilibrium. *Eur. J. Hum. Genet.* 17 (11), 1495–1500.
- Zimmer-Gembeck, M.J., Hunter, T.A., Waters, A.M., Pronk, R., 2009. Depression as a longitudinal outcome and antecedent of preadolescents' peer relationships and peer-relevant cognition. *Dev. Psychopathol.* 21 (2), 555–577.